AQA A-Level Further Mathematics (7367) — Topic 1.4

1.4: Series

1. Standard Summation Results

The three fundamental summation formulae (given in AQA formula booklet):

$$\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$$

$$\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2$$

Also useful: $\sum_{r=1}^{n} 1 = n$ and $\sum_{r=1}^{n} c = cn$ for constant c.

Example: Find $\sum_{r=1}^{n}(3r^2 - 2r + 1)$.

$$= 3 \cdot \frac{n(n+1)(2n+1)}{6} - 2 \cdot \frac{n(n+1)}{2} + n$$
$$= \frac{n(n+1)(2n+1)}{2} - n(n+1) + n$$
$$= \frac{n}{2}\left[(n+1)(2n+1) - 2(n+1) + 2\right]$$
$$= \frac{n}{2}(2n^2 + 3n + 1 - 2n - 2 + 2) = \frac{n}{2}(2n^2 + n + 1) = \frac{n(2n^2 + n + 1)}{2}$$

Example: Find $\sum_{r=n+1}^{2n} r^2$.

$$= \sum_{r=1}^{2n} r^2 - \sum_{r=1}^{n} r^2 = \frac{2n(2n+1)(4n+1)}{6} - \frac{n(n+1)(2n+1)}{6}$$
$$= \frac{n(2n+1)}{6}\left[2(4n+1) - (n+1)\right] = \frac{n(2n+1)(7n+1)}{6}$$

2. Method of Differences

If a sum can be written as $\sum_{r=1}^{n}[f(r) - f(r-1)]$ or $\sum_{r=1}^{n}[f(r) - f(r+1)]$, then the sum telescopes — most terms cancel.

Telescoping principle:

$$\sum_{r=1}^{n}[f(r) - f(r+1)] = f(1) - f(n+1)$$

$$\sum_{r=1}^{n}[f(r+1) - f(r)] = f(n+1) - f(1)$$

Example: Using partial fractions, find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$.

Partial fractions: $\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}$

$$\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+1}\right) = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right)$$
$$= 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

Example: Show that $\frac{1}{r(r+2)} = \frac{1}{2}\left(\frac{1}{r} - \frac{1}{r+2}\right)$, then find $\sum_{r=1}^{n} \frac{1}{r(r+2)}$.

Partial fractions confirm the identity.

$$\sum_{r=1}^{n} \frac{1}{r(r+2)} = \frac{1}{2}\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+2}\right)$$

Listing terms (write out first and last few):

$$\frac{1}{2}\left[\left(\frac{1}{1} - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n+1}\right) + \left(\frac{1}{n} - \frac{1}{n+2}\right)\right]$$

After cancellation: $= \frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{3}{4} - \frac{1}{2}\left(\frac{1}{n+1} + \frac{1}{n+2}\right)$

AQA Exam Tip: In method of differences, always write out at least the first three and last two terms explicitly to show the cancellation pattern. AQA mark schemes award marks specifically for showing the surviving terms.

3. Proof by Mathematical Induction

Structure of an Induction Proof

1. **Base case:** Show the statement is true for $n = 1$ (or the stated starting value).
2. **Inductive hypothesis:** Assume the statement is true for $n = k$ (state this clearly).
3. **Inductive step:** Prove the statement is true for $n = k + 1$, using the hypothesis.
4. **Conclusion:** "By the principle of mathematical induction, the result holds for all integers $n \geq 1$."

Induction for Summation

Example: Prove by induction that $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$.

Base case: $n = 1$: LHS $= 1$, RHS $= \dfrac{1 \cdot 2}{2} = 1$. True. ✓

Assume true for $n = k$: $\displaystyle\sum_{r=1}^{k} r = \frac{k(k+1)}{2}$

Show true for $n = k + 1$:
$$\sum_{r=1}^{k+1} r = \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}$$
This is the formula with $n = k + 1$. ✓

Conclusion: By mathematical induction, the result holds for all $n \geq 1$.

Induction for Divisibility

Example: Prove by induction that $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Base case: $n = 1$: $7^1 - 1 = 6 = 6 \times 1$. Divisible by 6. ✓

Assume true for $n = k$: $7^k - 1 = 6m$ for some integer m, i.e. $7^k = 6m + 1$.

Show true for $n = k + 1$:
$$7^{k+1} - 1 = 7 \cdot 7^k - 1 = 7(6m+1) - 1 = 42m + 7 - 1 = 42m + 6 = 6(7m+1)$$
This is divisible by 6. ✓

Conclusion: By mathematical induction, $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Induction for Matrix Powers

Example: Prove by induction that $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n$ for all $n \geq 1$.

Base case: $n = 1$: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. True ✓

Assume true for $n = k$: $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$

Show true for $n = k + 1$:
$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{k+1} = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$. ✓

AQA Exam Tip: The conclusion sentence is mandatory for full marks in AQA induction proofs. Do not omit it. The phrase "by the principle of mathematical induction" must appear.

4. Maclaurin Series

Maclaurin series: If $f(x)$ is infinitely differentiable at $x = 0$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

Standard Maclaurin Series

(These are given in the AQA formula booklet.)

Function	Maclaurin series	Valid for
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	All x
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + \frac{(-1)^{n+1}x^n}{n} + \cdots$	$-1 < x \leq 1$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	All x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	All x

$(1+x)^n$	$1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\cdots$	$\lvert x\rvert<1$ if $n\notin\mathbb{Z}^+$

Example: Find the Maclaurin series for $f(x)=\cos x$ up to and including the term in x^4, deriving it from first principles.

$f(x)=\cos x$: $f(0)=1$
$f'(x)=-\sin x$: $f'(0)=0$
$f''(x)=-\cos x$: $f''(0)=-1$
$f'''(x)=\sin x$: $f'''(0)=0$
$f^{(4)}(x)=\cos x$: $f^{(4)}(0)=1$

$$\cos x=1+0+\frac{-1}{2!}x^2+0+\frac{1}{4!}x^4+\cdots=1-\frac{x^2}{2}+\frac{x^4}{24}-\cdots$$

Combining Standard Series

Example: Find the first four non-zero terms of the Maclaurin series for $e^x\cos x$.

$$e^x=1+x+\frac{x^2}{2}+\frac{x^3}{6}+\cdots$$
$$\cos x=1-\frac{x^2}{2}+\frac{x^4}{24}+\cdots$$

Multiply term by term (collecting up to x^3):
Constant: 1; x: 1; x^2: $\frac{1}{2}-\frac{1}{2}=0$; x^3: $\frac{1}{6}-\frac{1}{2}=-\frac{1}{3}$
$e^x\cos x=1+x-\frac{x^3}{3}+\cdots$

5. Taylor Series

Taylor series about $x = a$:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Or equivalently, writing $x = a + h$:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots$$

Example: Find the Taylor expansion of $\ln x$ about $x = 1$, up to the term in $(x-1)^3$.

$f(x) = \ln x$: $f(1) = 0$
$f'(x) = 1/x$: $f'(1) = 1$
$f''(x) = -1/x^2$: $f''(1) = -1$
$f'''(x) = 2/x^3$: $f'''(1) = 2$

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \cdots$$

Example: Find the Maclaurin series for $\sin(x + \frac{\pi}{4})$ by using Taylor's expansion.

Let $a = \pi/4$, expand $\sin x$ about $\pi/4$... or substitute $x + \pi/4$ directly into the $\sin$ series using $\sin(A+B)$ and the series for $\sin u$ and $\cos u$.

$$\sin\left(x + \frac{\pi}{4}\right) = \sin x \cos\frac{\pi}{4} + \cos x \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}(\sin x + \cos x)$$
$$= \frac{1}{\sqrt{2}}\left[\left(x - \frac{x^3}{6} + \cdots\right) + \left(1 - \frac{x^2}{2} + \cdots\right)\right] = \frac{1}{\sqrt{2}}\left(1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \cdots\right)$$

AQA Exam Tip: The validity (range of x) of a combined series is the intersection of the validity ranges of its components. AQA may ask you to state this. For $\ln(1+x)$, validity is $-1 < x \leq 1$; the endpoint $x = 1$ is included (the series converges to $\ln 2$).

6. Summary of Key Results

Result	Formula
$\sum r$	$\dfrac{n(n+1)}{2}$
$\sum r^2$	$\dfrac{n(n+1)(2n+1)}{6}$
$\sum r^3$	$\left[\dfrac{n(n+1)}{2}\right]^2$
Telescoping	$\sum[f(r)-f(r+1)] = f(1)-f(n+1)$
Maclaurin	$f(x) = \sum_{n=0}^{\infty} \dfrac{f^{(n)}(0)}{n!}x^n$
Taylor about a	$f(x) = \sum_{n=0}^{\infty} \dfrac{f^{(n)}(a)}{n!}(x-a)^n$

7. Practice Checklist

- Can you use $\sum r$, $\sum r^2$, $\sum r^3$ to evaluate finite sums?
- Can you split a sum over $[n+1, 2n]$ using the full sum minus the first part?
- Can you use partial fractions and the method of differences to evaluate telescoping sums?
- Can you construct a complete proof by induction for summation, divisibility, and matrix power results?
- Can you derive and use the Maclaurin series for e^x, $\sin x$, $\cos x$, $\ln(1+x)$, and $(1+x)^n$?
- Can you find a Taylor series about a general point $x = a$?

Proof by Induction — Detailed Examples

Divisibility induction

Prove that $7^n - 1$ is divisible by 6 for all $n \geq 1$. Base case ($n = 1$): $7^1 - 1 = 6$, which is divisible by 6. ✓ **Inductive hypothesis:** Assume $7^k - 1 = 6m$ for some integer m. **Inductive step:** Show the result holds for $n = k + 1$:

$$7^{k+1} - 1 = 7 \cdot 7^k - 1 = 7(6m + 1) - 1 = 42m + 7 - 1 = 42m + 6 = 6(7m + 1)$$

This is divisible by 6. ✓ **Conclusion:** By the principle of mathematical induction, $7^n - 1$ is divisible by 6 for all $n \geq 1$.

Matrix power induction

Prove $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{pmatrix}$ **for all** $n \geq 1$. **Base ($n = 1$):** LHS $= \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$, RHS $= \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$. ✓ **Hypothesis:** Assume $M^k = \begin{pmatrix} 2^k & 2^k - 1 \\ 0 & 1 \end{pmatrix}$. **Step:** $M^{k+1} = M^k \cdot M = \begin{pmatrix} 2^k & 2^k - 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2^{k+1} & 2^k + (2^k - 1) \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2^{k+1} & 2^{k+1} - 1 \\ 0 & 1 \end{pmatrix}$. ✓

Induction structure (every time): Base case → State hypothesis for $n = k$ → Add/multiply to reach $n = k + 1$ → Conclude by PMI. Always write the conclusion sentence.

Maclaurin Series — Products and Quotients

When we need the Maclaurin series of a product $f(x)g(x)$, multiply the known series term by term. Only keep terms up to the required degree.

Find the series for $e^x \cos x$ up to x^4.

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots, \quad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$$

Multiply, keeping only terms up to degree 4:

$$e^x \cos x = 1 + x + \frac{x^2}{2} - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^3}{2} + \frac{x^4}{24} - \frac{x^4}{4} + \frac{x^4}{24} + \cdots$$
$$= 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \cdots$$

Quotient: Find the series for $\tan x$ up to x^5 using $\sin x / \cos x$. Perform the division $\sin x \div \cos x$:

$$(x - \tfrac{x^3}{6} + \tfrac{x^5}{120}) \div (1 - \tfrac{x^2}{2} + \tfrac{x^4}{24})$$

Using long division: $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$

Taylor Series — Worked Examples

The Taylor series of $f(x)$ about $x = a$ is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

Find the Taylor series of $\cos x$ about $x = \pi/2$ up to the $(x - \pi/2)^3$ term. $f(x) = \cos x$: $f(\pi/2) = 0$, $f'(\pi/2) = -\sin(\pi/2) = -1$, $f''(\pi/2) = -\cos(\pi/2) = 0$, $f'''(\pi/2) = \sin(\pi/2) = 1$.

$$\cos x = -(x - \tfrac{\pi}{2}) + 0 + \frac{1}{6}(x - \tfrac{\pi}{2})^3 - \cdots$$

Find the Taylor series of $\ln x$ about $x = 1$ up to $(x-1)^3$. $f(1) = 0$, $f'(1) = 1$, $f''(1) = -1$, $f'''(1) = 2$.

$$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \cdots$$

Error Estimation in Series Approximations

For an alternating series $\sum(-1)^n a_n$ where a_n is decreasing to zero, the error after n terms is bounded by the first omitted term $|a_{n+1}|$ (the **alternating series estimation theorem**).

Example: Use $\ln(1+x) = x - x^2/2 + x^3/3 - \cdots$ to estimate $\ln(1.2)$ correct to 3 decimal places. At $x = 0.2$: $0.2 - 0.02 + 0.00267 - 0.0004 + \ldots$ First omitted term (5th term): $0.2^5/5 = 0.000064$, which is less than 0.0001. $\ln(1.2) \approx 0.2 - 0.02 + 0.00267 - 0.0004 = 0.18227 \approx 0.182$.

AQA Exam Technique — Series

Key tips:

- **Induction:** The conclusion sentence is compulsory for full marks: "By the principle of mathematical induction, the result holds for all $n \geq 1$."
- **Maclaurin series:** State the range of validity if asked. For $\ln(1+x)$ it is $-1 < x \leq 1$; for $(1+x)^n$ it is $|x| < 1$.
- **Taylor series:** Compute derivatives at the centre point $x = a$, not at $x = 0$.
- **Limits via series:** If asked to evaluate a limit as $x \to 0$, expand the numerator and denominator separately, then cancel the lowest power.
- **Standard results:** You must know $\sum r = n(n+1)/2$, $\sum r^2 = n(n+1)(2n+1)/6$, $\sum r^3 = [n(n+1)/2]^2$ without derivation.

Series	Expansion	Valid for
e^x	$1 + x + x^2/2! + x^3/3! + \cdots$	All x
$\sin x$	$x - x^3/6 + x^5/120 - \cdots$	All x
$\cos x$	$1 - x^2/2 + x^4/24 - \cdots$	All x
$\ln(1+x)$	$x - x^2/2 + x^3/3 - \cdots$	$-1 < x \leq 1$
$(1+x)^n$	$1 + nx + n(n-1)x^2/2! + \cdots$	$\|x\| < 1$ (non-integer n)

Method of Differences — Further Examples

The method of differences works when each term can be written as $f(r+1) - f(r)$ or $f(r) - f(r+k)$ for some function f. The sum then telescopes.

Example 1: Find $\sum_{r=1}^{n} \frac{2}{r(r+2)}$. Partial fractions: $\frac{2}{r(r+2)} = \frac{1}{r} - \frac{1}{r+2}$. This is a "gap-2" telescoping. Writing out the first few and last few terms:

$$\left(1 - \frac{1}{3}\right) + \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{5}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+2}\right)$$

Terms that survive: $\frac{1}{1} + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2} = \frac{3}{2} - \frac{1}{n+1} - \frac{1}{n+2}$.

Example 2: Use the identity $r(r+1)(r+2) - (r-1)r(r+1) = 3r(r+1)$ to find $\sum_{r=1}^{n} r(r+1)$.

$$3\sum_{r=1}^{n} r(r+1) = \sum_{r=1}^{n} [r(r+1)(r+2) - (r-1)r(r+1)] = n(n+1)(n+2) - 0$$

$$\sum_{r=1}^{n} r(r+1) = \frac{n(n+1)(n+2)}{3}$$

Summation Formulae — Derivation of $\sum r^3$

Prove $\sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2$ **by induction. Base:** $n = 1$: LHS=1, RHS=1. ✓ **Hypothesis:** $\sum_{r=1}^{k} r^3 = \frac{k^2(k+1)^2}{4}$. **Step:** Add $(k+1)^3$:

$$\frac{k^2(k+1)^2}{4} + (k+1)^3 = \frac{(k+1)^2}{4}[k^2 + 4(k+1)] = \frac{(k+1)^2(k+2)^2}{4}$$

This is the formula with $n = k+1$. ✓

Using Maclaurin Series to Approximate Integrals

Approximate $\int_0^{0.3} e^{-x^2}\,dx$ to 4 d.p. Replace x by $-x^2$ in $e^x = 1 + x + x^2/2 + \cdots$:

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} + \cdots$$

Integrate term by term:

$$\int_0^{0.3} \left(1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6}\right) dx = \left[x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42}\right]_0^{0.3}$$

$$= 0.3 - 0.009 + 0.000243 - 0.0000052 \approx 0.2911$$

Error bound: next term $\approx 0.3^9/216 \approx 5 \times 10^{-8}$. Accurate to 4 d.p.

Binomial Series — Applications

Example: Find the first three terms of $\dfrac{1}{\sqrt{4+x}}$ for small x.

$$\frac{1}{\sqrt{4+x}} = \frac{1}{2}(1 + x/4)^{-1/2} = \frac{1}{2}\left[1 + \left(-\frac{1}{2}\right)\frac{x}{4} + \frac{(-1/2)(-3/2)}{2} \cdot \frac{x^2}{16} + \cdots\right]$$

$$= \frac{1}{2}\left[1 - \frac{x}{8} + \frac{3x^2}{128} + \cdots\right] = \frac{1}{2} - \frac{x}{16} + \frac{3x^2}{256} + \cdots$$

Valid for $|x/4| < 1$, i.e.\ $|x| < 4$.

Standard Sum Results Summary

Sum	Formula
$\sum_{r=1}^{n} 1$	n

$\sum_{r=1}^{n} r$	$\frac{n(n+1)}{2}$
$\sum_{r=1}^{n} r^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{r=1}^{n} r^3$	$\left[\frac{n(n+1)}{2}\right]^2$
$\sum_{r=1}^{n} r(r+1)$	$\frac{n(n+1)(n+2)}{3}$
$\sum_{r=1}^{n} r(r+1)(r+2)$	$\frac{n(n+1)(n+2)(n+3)}{4}$

Proof by Induction — More Examples

Example 1 — Prove $\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$:

Base: $n = 1$: LHS$= 1$, RHS$= \frac{1 \cdot 2}{2} = 1$. ✓

Inductive step: Assume true for $n = k$. Then:

$$\sum_{r=1}^{k+1} r = \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1) + 2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

This is the formula for $n = k + 1$. By induction, true for all $n \geq 1$. ✓

Example 2 — Prove $7^n - 1$ is divisible by 6:

Base: $n = 1$: $7 - 1 = 6$, divisible by 6. ✓

Inductive step: Assume $7^k - 1 = 6m$ for some integer m. Then:

$7^{k+1} - 1 = 7 \cdot 7^k - 1 = 7(7^k - 1) + 7 - 1 = 7 \cdot 6m + 6 = 6(7m + 1)$.

This is divisible by 6. By induction, true for all $n \geq 1$. ✓

Maclaurin Series — Derivation from Scratch

The Maclaurin series is a Taylor series about $x = 0$:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

Derive the series for $\sin x$:
$f(x) = \sin x$: $f(0) = 0$. $f'(x) = \cos x$: $f'(0) = 1$. $f''(x) = -\sin x$: $f''(0) = 0$.
$f'''(x) = -\cos x$: $f'''(0) = -1$. $f^{(4)}(x) = \sin x$: $f^{(4)}(0) = 0$. Cycle repeats.
$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$

Derive series for $\ln(1+x)$:
$f(x) = \ln(1+x)$: $f(0) = 0$. $f'(x) = \frac{1}{1+x}$: $f'(0) = 1$.
$f''(x) = -\frac{1}{(1+x)^2}$: $f''(0) = -1$. $f'''(x) = \frac{2}{(1+x)^3}$: $f'''(0) = 2$.
In general: $f^{(n)}(0) = (-1)^{n+1}(n-1)!$.
\(\ln(1+x) = x-\dfrac{x^2}{2}+\dfrac{x^3}{3}-\dfrac{x^4}{4}+\cdots\quad(-1

Taylor Series and Approximations

Taylor series about $x = a$:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

Linear approximation: $f(x) \approx f(a) + f'(a)(x-a)$ (tangent line at a).
Quadratic approximation: Include the second-order term for better accuracy.

Example — Approximate $\sqrt{1.1}$:
Let $f(x) = \sqrt{x}$, $a = 1$. $f(1) = 1,\ f'(1) = \frac{1}{2},\ f''(1) = -\frac{1}{4}$.
$\sqrt{1.1} \approx 1 + \frac{1}{2}(0.1) - \frac{1}{8}(0.1)^2 = 1 + 0.05 - 0.00125 = 1.04875$.
(Exact: $1.04881\ldots$)

Method of Differences — Full Examples

Example 1: Find $\sum_{r=1}^{n} \frac{1}{r(r+1)}$.

Write $\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}$ (partial fractions).

Telescoping: $\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1} = \frac{n}{n+1}$.

Example 2: Find $\sum_{r=1}^{n} r(r+1)(r+2) - r(r+1)(r+2)(r+3)$... actually, use a difference form.

Alternative: Sum $\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)}$.

$\frac{1}{(2r-1)(2r+1)} = \frac{1}{2}\left(\frac{1}{2r-1} - \frac{1}{2r+1}\right)$.

Sum $= \frac{1}{2}\left(1 - \frac{1}{2n+1}\right) = \frac{n}{2n+1}$.

Convergence and Limits of Series

A series $\sum_{r=1}^{\infty} a_r$ **converges** if the partial sums $S_n \to L$ as $n \to \infty$.

Geometric series: $\sum_{r=0}^{\infty} ar^n = \frac{a}{1-r}$ for $|r| < 1$; diverges for $|r| \geq 1$.

Necessary condition: If $\sum a_r$ converges, then $a_r \to 0$. But the converse fails (e.g. harmonic series $\sum 1/r$ diverges even though $1/r \to 0$).

Example — Sum to infinity: $\sum_{r=2}^{\infty} \frac{3}{r(r-1)}$.

$\frac{3}{r(r-1)} = 3\left(\frac{1}{r-1} - \frac{1}{r}\right)$. Sum from 2 to ∞:

$S = 3\left(1 - \lim_{r\to\infty} \frac{1}{r}\right) = 3 \cdot 1 = 3$.

Induction — Inequality Proof

Example — Prove $2^n > 2n$ for $n \geq 3$:
Base: $n = 3$: $8 > 6$. ✓
Inductive step: Assume $2^k > 2k$ for some $k \geq 3$. Then:
$2^{k+1} = 2 \cdot 2^k > 2 \cdot 2k = 4k = 2(k+1) + 2(k-1) > 2(k+1)$ since $k \geq 3 > 1$.
So $2^{k+1} > 2(k+1)$. By induction, true for all $n \geq 3$. ✓

Proof by Induction — Divisibility Examples

Structure of an induction proof:
1. *Basis step:* Verify the statement for the smallest value (usually $n = 1$ or $n = 0$).
2. *Inductive hypothesis:* Assume the statement holds for $n = k$.
3. *Inductive step:* Using the assumption, prove it holds for $n = k + 1$.
4. *Conclusion:* By the principle of mathematical induction, the result holds for all $n \geq n_0$.

Example — Prove $5^n - 1$ is divisible by 4 for all $n \geq 1$:
Base: $n = 1$: $5^1 - 1 = 4 = 4 \times 1$. ✓
Hypothesis: Assume $4 \mid 5^k - 1$, i.e. $5^k - 1 = 4m$ for some integer m.
Step: $5^{k+1} - 1 = 5 \cdot 5^k - 1 = 5(4m+1) - 1 = 20m + 4 = 4(5m+1)$.
So $4 \mid 5^{k+1} - 1$. By induction, true for all $n \geq 1$. ✓

Example — Prove $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 3^n & \frac{3^n-1}{2} \\ 0 & 1 \end{pmatrix}$ for $n \geq 1$:
Base: $n = 1$: LHS$= \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$, RHS$= \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$. ✓
Step: Assume true for $n = k$. Multiply by $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$:
$$\begin{pmatrix} 3^k & \frac{3^k-1}{2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3^{k+1} & 3^k + \frac{3^k-1}{2} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3^{k+1} & \frac{3^{k+1}-1}{2} \\ 0 & 1 \end{pmatrix}. \checkmark$$

For divisibility proofs, write $f(k+1) = \alpha \cdot f(k) + (\text{multiple of } d)$ to make it clear d divides $f(k+1)$. Do not just say "therefore divisible" — show the algebra.

Maclaurin Series — Combining Series

You can multiply, divide, substitute into, and add known Maclaurin series to get new ones without differentiating from scratch.

Example 1 — Find the Maclaurin series for $e^x \cos x$ up to x^3:
$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$
$\cos x = 1 - \frac{x^2}{2} + \cdots$
Multiply (keeping terms up to x^3):
$e^x \cos x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)\left(1 - \frac{x^2}{2}\right) + \cdots$
$= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^2}{2} - \frac{x^3}{2} + \cdots = 1 + x - \frac{x^3}{3} + \cdots$

Example 2 — Find the Maclaurin series for $\sin(\ln(1+x))$ up to x^2:
Let $u = \ln(1+x) = x - \frac{x^2}{2} + \cdots$
$\sin u = u - \frac{u^3}{6} + \cdots \approx u$ to order x^2.
So $\sin(\ln(1+x)) \approx x - \frac{x^2}{2} + \cdots$

When finding series for composite functions, substitute the inner series into the outer one. Only keep terms up to the required power — drop any term with degree higher than needed early to reduce working.

Taylor Series — Further Examples

Example — Expand $\tan x$ about $x = 0$ up to x^5:
$f(x) = \tan x$: $f(0) = 0$, $f'(0) = \sec^2 0 = 1$, $f''(0) = 2\sec^2 0 \tan 0 = 0$,

$f'''(0) = 2$, $f^{(4)}(0) = 0$, $f^{(5)}(0) = 16$.

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$$

Example — Expand $\ln \cos x$ about $x = 0$ up to x^4:

Use $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots$, so $\cos x = 1 + u$ where $u = -\frac{x^2}{2} + \frac{x^4}{24} - \cdots$

$\ln(1+u) = u - \frac{u^2}{2} + \cdots$

$u = -\frac{x^2}{2} + \frac{x^4}{24}$, $\frac{u^2}{2} = \frac{x^4}{8} + \cdots$

$\ln \cos x = -\frac{x^2}{2} + \frac{x^4}{24} - \frac{x^4}{8} + \cdots = -\frac{x^2}{2} - \frac{x^4}{12} + \cdots$

Error Estimation in Series

When you truncate a series, the **remainder** R_n is the sum of all omitted terms.

For alternating series: $|R_n| \leq |a_{n+1}|$ (the first omitted term gives an upper bound for the error).

For power series within their radius of convergence, use the Lagrange remainder:

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} x^{n+1} \quad \text{for some } c \text{ between } 0 \text{ and } x.$$

Example — Estimate the error in using $\sin x \approx x - \frac{x^3}{6}$ for $x = 0.1$:

Next term: $\frac{x^5}{120} = \frac{(0.1)^5}{120} = \dfrac{10^{-5}}{120} \approx 8.3 \times 10^{-8}$.

So the error is less than 10^{-7} — an excellent approximation.

Standard Results Summary

Series	Expansion	Validity
e^x	$1 + x + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots$	All x

$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$	All x
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$	All x
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$	$-1 < x \leq 1$
$(1+x)^n$	$1 + nx + \frac{n(n-1)}{2!}x^2 + \cdots$	$\lvert x \rvert < 1$ (non-integer n)
$\arctan x$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$	$\lvert x \rvert \leq 1$

AQA Exam Technique

Induction proofs: You must use the *exact* wording prescribed by AQA. The conclusion must state "by the principle of mathematical induction, the result is true for all $n \geq n_0$". Never skip the base case, and always write the inductive hypothesis explicitly.

Method of differences: Always write out at least 4 rows of the telescoping sum, then write the last 2 rows. State clearly which terms cancel and which survive. Don't forget to simplify the final answer.

Maclaurin series validity: AQA will often ask you to *state* the range of validity. For $\ln(1+x)$ the range is \(-1

> **Standard result sums:** The formulae $\sum r$, $\sum r^2$, $\sum r^3$ are given in the formula booklet. Always quote the formula, substitute, then simplify step by step showing the factorisation. Leaving an unexpanded polynomial will lose accuracy marks.